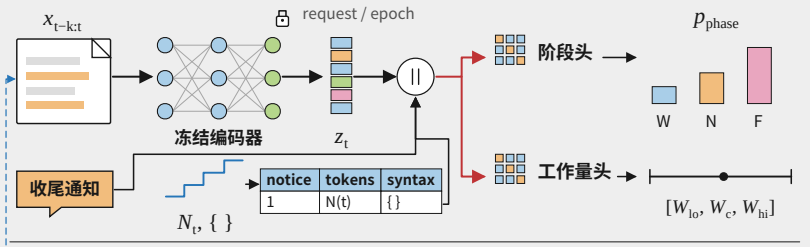
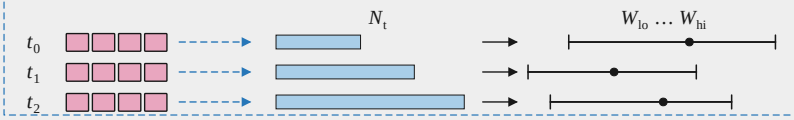
事件与内容驱动的执行需求预测

已送达正文 + 服务进度 → 阶段 / 工作量双头 → Runtime 决定启动时机

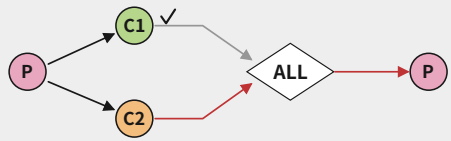
(a) 内容、事件与滚动剩余工作预测



在线快照 → 滚动修正工作端点

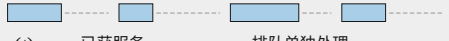


(b) JOIN 投影与已观测服务进度



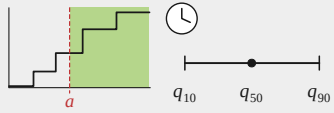
group	pending	signal
g	{C1, C2}	WAIT
g	{C2}	NOTICE

$$\tau_{join}(t) = \frac{[W - \Delta N]_+}{v(t)}$$ pending = 1



(c) 存活条件下的工具时机

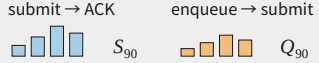
project	cmd	done
p	k	T1
p	k	T2



$$P(T \le a + r \mid T > a) = \frac{F(a+r) - F(a)}{1 - F(a)} \rightarrow \tau_{tool}$$

(d) Runtime：成本、时机与物理校验

dir	shape	bytes
H2D	extents	B



$$H = \min(L, S_{90} + Q_{90} + \Delta_{obs} + \varepsilon_{early})$$

